roam: Remote Objects with Active-binding Magic

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Data in R

```r
library(tsibble)
tourism
```

```
# A tsibble: 24,320 x 5 [1Q]
# Key:       Region, State, Purpose [304]
   Quarter Region   State           Purpose  Trips
     <qtr> <chr>    <chr>           <chr>    <dbl>
 1 1998 Q1 Adelaide South Australia Business  135.
 2 1998 Q2 Adelaide South Australia Business  110.
 3 1998 Q3 Adelaide South Australia Business  166.
 4 1998 Q4 Adelaide South Australia Business  127.
 5 1999 Q1 Adelaide South Australia Business  137.
 6 1999 Q2 Adelaide South Australia Business  200.
 7 1999 Q3 Adelaide South Australia Business  169.
 8 1999 Q4 Adelaide South Australia Business  134.
 9 2000 Q1 Adelaide South Australia Business  154.
10 2000 Q2 Adelaide South Australia Business  169.
# i 24,310 more rows
```

R package

Source package

```
tsibble/
├── DESCRIPTION
├── NAMESPACE
├── R/
├── data/
│   └── tourism.rda
├── data-raw/
│   ├── tourism_raw.csv
│   └── get_tourism.R
└── man/
```

Binary package

```
tsibble/
├── DESCRIPTION
├── NAMESPACE
├── INDEX
├── Meta/
├── R/
├── help/
└── data/
    ├── Rdata.rdb
    ├── Rdata.rds
    └── Rdata.rdx
```

Distribution

The Comprehensive R Archive Network (CRAN)

```r
install.packages("tsibble")
```

- Precompiled (binary) distribution of contributed packages.
- Currently hosts 2.3424×10^4 packages.

Github and alike

```r
remotes::install_github("FinYang/roam")
```

- Compiled locally from source.

Getting help

```
?tourism
```

Australian domestic overnight trips

Description:

```
A dataset containing the quarterly overnight trips from 1998 Q1 to
2016 Q4 across Australia.
```

Usage:

```
tourism
```

Format:

```
A tsibble with 23,408 rows and 5 variables:
```

List of data

```
data()
```

Table 1: Data sets

Package	Item	Title
tsibble	pedestrian	Pedestrian counts in the city of Melbourne
tsibbletourism		Australian domestic overnight trips
datasets	AirPassengers	Monthly Airline Passenger Numbers 1949-1960
datasets	BJsales	Sales Data with Leading Indicator
datasets	BJsales.lead	Sales Data with Leading Indicator

List of data in a package

```r
data(package = "tsibble")
```

Table 2: Data sets in tsibble

Item	Title
pedestrian	Pedestrian counts in the city of Melbourne
tourism	Australian domestic overnight trips

Limitations

Source package

```
tsibble/
├── DESCRIPTION
├── NAMESPACE
├── R/
├── data/
│   └── tourism.rda
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```

- Data included in package
- CRAN size limit 5 MB
  - Including documentation, vignettes, pictures

Limitations

Source package

```
tsibble/
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├── NAMESPACE
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│   └── tourism.rda
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│   ├── tourism_raw.csv
│   └── get_tourism.R
└── man/
```

- Data included in package
- CRAN size limit 5 MB
  - Including documentation, vignettes, pictures
- The data are updated only when the package is updated
  - CRAN recommended update frequency “No more than every 1-2 months”

Data with `roam`

```r
library(roam.demo)
options(roam.autodownload = TRUE)
bee
```

	X	year	months	state	colony_n	colony_max	colony_lost
1	1	2015	January-March	Alabama	7000	7000	1800
2	2	2015	January-March	Arizona	35000	35000	4600
3	3	2015	January-March	Arkansas	13000	14000	1500
4	4	2015	January-March	California	1440000	1690000	255000
5	5	2015	January-March	Colorado	3500	12500	1500
6	6	2015	January-March	Connecticut	3900	3900	870
7	7	2015	January-March	Florida	305000	315000	42000
8	8	2015	January-March	Georgia	104000	105000	14500
9	9	2015	January-March	Hawaii	10500	10500	380
10	10	2015	January-March	Idaho	81000	88000	3700
11	11	2015	January-March	Illinois	6000	10500	4200
12	12	2015	January-March	Indiana	9000	9500	2100

Help with `roam`

```
?bee
```

beeeeeeee

Description:

beeeeeeee

Usage:

```
bee
```

Format:

buzzzzzzzz

But storage with `roam`?

Source package

```
roam.demo/
├── DESCRIPTION
├── NAMESPACE
├── R/
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```

Binary package

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Binary package

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├── R/
└── help/
```

- The data is not combined with the package.

`roam`: Remote Objects with Active-binding Magic

1. Allow ‘regular looking’ R objects in packages to exceed the 5MB limit.
2. Support updating of datasets without updating packages by using functions that pull from remote resources.
3. Make it easy for packages to include these ‘roaming’ datasets.

Data with `roam`

```r
library(roam.demo)
options(roam.autodownload = TRUE)
bee
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	X	year	months	state	colony_n	colony_max	colony_lost
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Data with `roam`

```r
library(roam.demo)
bee
```

```
The roam data object "bee" in package roam.demo does not exist locally
```

Data with `roam`

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bee
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```
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```

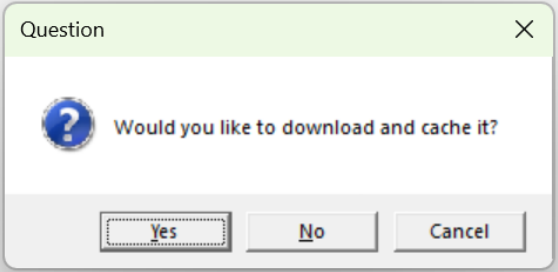

Data with `roam`

```
Data retrieved

        X year           months          state colony_n colony_max colony_lost
1       1 2015    January-March        Alabama     7000       7000        1800
2       2 2015    January-March        Arizona    35000      35000        4600
3       3 2015    January-March       Arkansas    13000      14000        1500
4       4 2015    January-March     California  1440000    1690000      255000
5       5 2015    January-March       Colorado     3500      12500        1500
6       6 2015    January-March    Connecticut     3900       3900         870
7       7 2015    January-March        Florida   305000     315000       42000
8       8 2015    January-March        Georgia   104000     105000       14500
9       9 2015    January-March         Hawaii    10500      10500         380
10     10 2015    January-March          Idaho    81000      88000        3700
11     11 2015    January-March       Illinois     6000      10500        4200
12     12 2015    January-March        Indiana     9000       9500        2100
13     13 2015    January-March           Iowa    12500      17500        3100
14     14 2015    January-March         Kansas     4600       7000        1600
15     15 2015    January-March       Kentucky     7500      10500        4100
16     16 2015    January-March      Louisiana    51000      51000        8000
```

Active bindings

```r
makeActiveBinding(
  "random_number",
  \() rnorm(1L),
  .GlobalEnv
)
```

```r
random_number
```

```
[1] -0.3380621
```

```r
random_number
```

```
[1] 0.9391643
```

```r
random_number
```

```
[1] 1.737719
```

Active bindings

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makeActiveBinding(
  "random_number",
  \() rnorm(1L),
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```r
random_number
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```
[1] -0.3380621
```

```r
random_number
```

```
[1] 0.9391643
```

```r
random_number
```

```
[1] 1.737719
```

- Re-computed every time they are accessed.

Evil: Stochastic truth

```r
T
```

```
[1] TRUE
```

```r
makeActiveBinding(
  "T",
  \() sample(c(TRUE, FALSE), 1, prob = c(0.95, 0.5)),
  .GlobalEnv
)
```

```r
T
```

```
[1] FALSE
```

```r
T
```

```
[1] TRUE
```

```r
T
```

```
[1] FALSE
```

Evil: Mathematical non-constant

```r
pi
```

```
[1] 3.141593
```

```r
makeActiveBinding(
  "pi",
  \() 3.14 + rnorm(1L, sd = 0.1),
  .GlobalEnv
)
```

```r
pi
```

```
[1] 2.986793
```

```r
pi
```

```
[1] 3.180207
```

```r
pi
```

```
[1] 3.28602
```

Names

Who

- **Developer, Pacakge writer**: The people who uses `roam` in their packages to provide data.
- **User**: The people who uses the packages developed by the *developer* and consumes the data.

Names

Who

- **Developer, Pacakge writer**: The people who uses `roam` in their packages to provide data.
- **User**: The people who uses the packages developed by the *developer* and consumes the data.

What

- `roam`: The package that provide the tools.
- `roam.demo`: A demo package that uses `roam` to provide data.

Developer without roam

```r
# data-raw/bee.R
library(readr)
bee <- read_csv(
  "data-raw/bee.csv"
)
usethis::use_data(bee)
```

Source package

```
roam.demo/
├── DESCRIPTION
├── NAMESPACE
├── R/
├── data/
│   └── bee.rda
├── data-raw/
│   ├── bee.csv
│   └── bee.R
└── man/
```

Developer with roam

Definition

```r
obtainer <- function(version) {
  readr::read_csv("path/to/bee.csv")
}
bee <- roam::new_roam('roam.demo', 'bee', obtainer)
```

Activation

```r
.onLoad <- function(libname, pkgname) {
  roam::roam_activate(bee)
}
```

Two-step process

`roam` objects are regular looking objects that works like functions.

1. **Definition**: Create the function
2. **Activation**: Turn it into a `roam` object.

1. Definition

```
new_roam(package, name, obtainer, ...)
```

- `package`: the name of the package as a string
- `name`: the name of the roam object. It should be the same as the name to which the roam object is assigned.
- `obtainer`: the function the developer defines to retrieve the dataset from the remote source.

Obtainer

```
obtainer <- function(version) {
  readr::read_csv(
    "https://raw.githubusercontent.com/finyang/roam/master
  )
}
```

- `obtainer`: the function the developer defines to retrieve the dataset from the remote source.
- Needs to have (at least) one argument called `version` for versioning purpose.

Functional programming: Function factory

Function factory

A function that makes functions

```
obtainer <- function(version) readr::read_csv("url/to/csv"
bee <- roam::new_roam('roam.demo', 'bee', obtainer)
str(bee)
```

```
function (...)
 - attr(*, "class")= chr "roam_object"
 - attr(*, "package")= chr "pkg"
 - attr(*, "name")= chr "bee"
```

2. Activation

```
.onLoad <- function(libname, pkgname) {
  roam::roam_activate(bee)
}
```

- Active bindings are not preserved during package installation.
- `.onLoad` is called when the package `roam.demo` is loaded
  - `roam_objects` are activated during package loading

So where is the data?

User-specific data directory

- win: C:/Users/P70~0/AppData/Local/r-roam/r-roam/
- mac: ~/Library/Application Support/r-roam/
- unix: ~/.local/share/r-roam/

```
r-roam/
├── roam.demo/
│   ├── tidytuesday2026Jan.RData
│   ├── tidytuesday2026Jan.txt
│   ├── bee.RData
│   └── bee.txt
...
```

Challenges

How is this possible
FinYang committed 16 hours ago · ✕ 4 / 5

Try again
FinYang committed 16 hours ago · ✕ 4 / 5

Try again
FinYang committed 16 hours ago · ✕ 0 / 5

Try again without loading namespace
FinYang committed 16 hours ago · ✕ 0 / 5

Try lots of prints
FinYang committed 17 hours ago · ✕ 0 / 5

Try to set libpath in check()
FinYang committed 17 hours ago · ✕ 0 / 5

Try to find MASS in test
FinYang committed 17 hours ago · ✕ 4 / 5

Print Library in test
FinYang committed 18 hours ago · ✕ 4 / 5

Can’t reach active bindings

Features

```
# Install a specific version
roam_install(bee, version = "1.2.1")
# Install the latest version
roam_update(bee)
# Get the current version
roam_version("roam.demo", "bee")
# Delete the local cache
roam_delete(bee)
```

Can’t reach active bindings

Features

```r
# Install a specific version
roam_install(bee, version = "1.2.1")
# Install the latest version
roam_update(bee)
# Get the current version
roam_version("roam.demo", "bee")
# Delete the local cache
roam_delete(bee)
```

But the `roam_object` is not accessible after activtation, only the output

```r
class(bee)
```

```
[1] "data.frame"
```

```r
typeof(bee)
```

```
[1] "list"
```

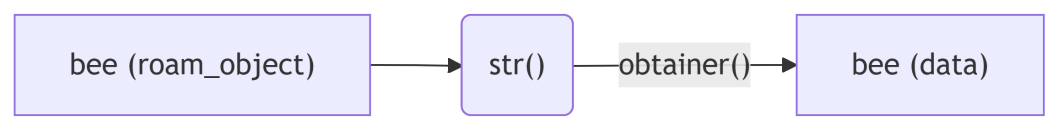
```r
str(bee)
```

```
'data.frame':   1222 obs. of  11 variables:
 $ X             : int  1 2 3 4 5 6 7 8 9
 $ year          : int  2015 2015 2015 201
 $ months        : chr  "January-March" "J
 $ state         : chr  "Alabama" "Arizona
 $ colony_n      : int  7000 35000 13000 1
```

Evaluation order

```
str(bee)
```

It would work if



But it is actually

Solution: flags (something like environment variable)

```
roam_delete(bee)
```

```
# roam_delete
{
  roam_flag$delete <- TRUE
  bee
  roam_flag$delete <- FALSE
}
```

```
# bee
{
  if (roam_flag$delete) {
    # delete cache
  }
  if (roam_flag$update) {
    # update bee
  }
  ...
}
```

Versioning

- Cannot input or output version to or from `roam_object`
  - The only output should be the data itself.
  - Should not modify the data from `roam`
  - Cannot attach version info to the data

Versioning

- Cannot input or output version to or from `roam_object`
  - The only output should be the data itself.
  - Should not modify the data from `roam`
  - Cannot attach version info to the data
- User supply version and developer version are not the same
  - User says `latest` while developer says `2026-03-26`.
  - Developer should have control.

Internals

```r
roam_install(bee, version = "latest")
```

```r
# roam_install
roam_flag$install <- TRUE
roam_flag$version <- "latest"
bee
roam_flag$install <- FALSE
roam_flag$version <- NA
```

```r
# bee
if (roam_flag$install) {
  obtainer(
    roam_flag$version
  )
}
```

Developer has control

```r
obtainer <- function(version) {
  if (version == "latest") {
    developer_version <- "0.1.0"
  }

  # Tell roam what is the version to remember
  roam_set_version(developer_version)

  # Specify how to download based on version
  return(download_with(version = version))
}
```

Remember version

```
r-roam/
├── roam.demo/
│   ├── tidytuesday2026Jan.RData
│   ├── tidytuesday2026Jan.txt
│   ├── bee.RData
│   └── bee.txt
...
```

- The version is stored in a separate `.txt` file in the same folder as the data.
- `roam` never touches the data provided by the developer.

Coauthor

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Will be visiting us on 18–19 June

Contact

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roam: github.com/FinYang/roam

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